PART 5

Expert Technical Documentation

Document ID: EXP-PLC-005 | Revision: 1.0 | Status: SYNTHETIC DEMO

Audience: controls engineers and senior maintenance staff. Content summarises key behaviour of the S7-1200 from the system manual (04/2012, firmware V3.0) as applied to the Filler Cell. Page references are to that manual.

1. System architecture

The Filler Cell uses a CPU 1214C: 75 KB work memory, 4 MB load memory, 10 KB retentive memory, 14 digital inputs and 10 digital outputs on board, 2 analog inputs, up to 8 signal modules, one signal board slot (signal, communication or battery board) and up to 3 communication modules on the left side. It has a single PROFINET port, 3 HMI connections, and 4 pulse outputs. Boolean execution is about 0.08 microseconds per instruction.

Item	Filler Cell configuration (synthetic)
CPU	1214C DC/DC/DC, firmware V3.0
Right side (SM)	SM 1223 DI/DQ 16x24VDC; SM 1231 AI 4 x 16 bit
Signal board slot	BB 1297 battery board (RTC backup)
Left side (CM)	None
HMI	KTP 700-class panel over PROFINET (generic)
Programming	STEP 7 (TIA Portal), LAD and SCL

2. CPU operating modes and execution

The CPU has three operating modes: STOP, STARTUP and RUN. In STOP the program is not executed and downloads are possible. In STARTUP, startup OBs (OB 100 and others) run once and the process image is initialised. In RUN the scan cycle repeats: read inputs into the process image, execute the program cycle OB (OB 1), write outputs, then communication and self-test tasks. Interrupt OBs can pre-empt the cycle according to priority. The scan cycle is monitored by a watchdog (maximum cycle time) which raises time-error OB 80; if exceeded again the CPU goes to STOP. Outputs on a RUN-to-STOP transition are configurable (substitute value or freeze last value).

3. Status LEDs - quick diagnostic reference

LED	State	Meaning
RUN/STOP	Solid yellow	CPU in STOP
RUN/STOP	Solid green	CPU in RUN
RUN/STOP	Alternating yellow/green	CPU in STARTUP
ERROR	Flashing red	Error such as internal fault, memory card error or configuration error
ERROR	Solid red	Hardware defect
MAINT	Flashing yellow	Maintenance requested
LINK / Rx-Tx	Green / flashing yellow	PROFINET link up / data activity

Always combine LED state with the diagnostic buffer before acting. See manual section 14.1 for the full LED table.

4. Memory and memory card

Load memory holds the project (program, data, configuration); work memory holds the code and data used at run time; retentive memory preserves selected data through power loss. A SIMATIC memory card can act as program card (the CPU runs from the card), transfer card (copies the project to internal load memory) or firmware update card. Removing a program card while powered causes a CPU STOP and error, as seen in

WO-26-0742. Avoid handling cards with power on.

5. Communication

Function	Notes
PROFINET	IP address and device name must be unique. The CPU supports open user communication (TCP, ISO on TCP, UDP), S7 communication (GET/PUT), and IO-device connections.
Web server	Standard pages give identification, diagnostic buffer, module information, variable status, data logs and firmware update. Enable only if needed and restrict access.
Serial / Modbus	Requires CM 1241 or CB 1241 for PtP, USS and Modbus RTU; Modbus TCP uses the PROFINET port with MB_CLIENT / MB_SERVER.

6. Protection and security

- **Access protection**: password levels restrict full access, read access and HMI access to the CPU.
- **Know-how protection** hides block code. **Copy protection** binds the program to a specific memory card or CPU serial number.
- **Lost password recovery** requires resetting the CPU, which erases the program, so keep current backups.
- Treat the web server and PROFINET as networked attack surfaces. Segment the control network and avoid exposing the CPU to the office network.

7. Diagnostics and troubleshooting guide

Symptom	Likely cause	Action
CPU in STOP, ERROR flashing red	Memory card issue, configuration error or program error	Read the diagnostic buffer; check card seating; compare project online/offline; escalate to L2
CPU in STOP after long scan	Cycle time exceeded watchdog (OB 80 event)	Check cycle monitoring; review loops and communication blocks; review high interrupt load
Intermittent HMI or device loss	Cabling, switch port, duplicate IP or device name	Check LINK LED, patch cable, ping; verify names/IPs; review diagnostic buffer
Analog value noisy or drifting	Loose terminal, wrong wiring, scaling error, EMI	Check terminals, shield grounding and module parameters; verify scaling with a source
Time of day drifts after outage	RTC backup exhausted (about 20 days typical without battery board)	Check BB 1297 and set clock; consider NTP synchronisation
Output not switching	Forced value, output disabled in STOP, module fault or load fault	Check force table, CPU mode, module LEDs and wiring

8. Power budget and expansion rules

Each CPU supplies a 5 V DC logic budget to its signal modules, boards and communication modules. Before adding a module, calculate the power budget (manual Appendix B) and confirm the total 5 V consumption does not exceed the CPU's supply and that 24 V sensor-supply loading stays within limits. Do not exceed the maximum of 8 SMs on the 1214C or 3 CMs.

9. Change management and best practice

- Use version control or dated project archives; one archive per change, with a short change note.
- Use download-in-RUN only after checking its prerequisites; otherwise download in STOP during a planned window.
- Name tags and blocks consistently; keep scaling equations (for example normalised analog scaling) in reusable function blocks.
- Test changes offline or on a spare CPU before production deployment.
- Review the diagnostic buffer regularly. It is a rolling log and older entries are lost.

10. Glossary

Term	Meaning
CPU	Central processing unit of the PLC
SM / SB / CB / BB	Signal module / signal board / communication board / battery board
CM / CP	Communication module / communication processor
OB / FB / FC / DB	Organization block / function block / function / data block
PROFINET	Industrial Ethernet standard used by the CPU's built-in port
LOTO	Lockout/tagout energy isolation

End of documentation pack. All content is synthetic and for demonstration only.